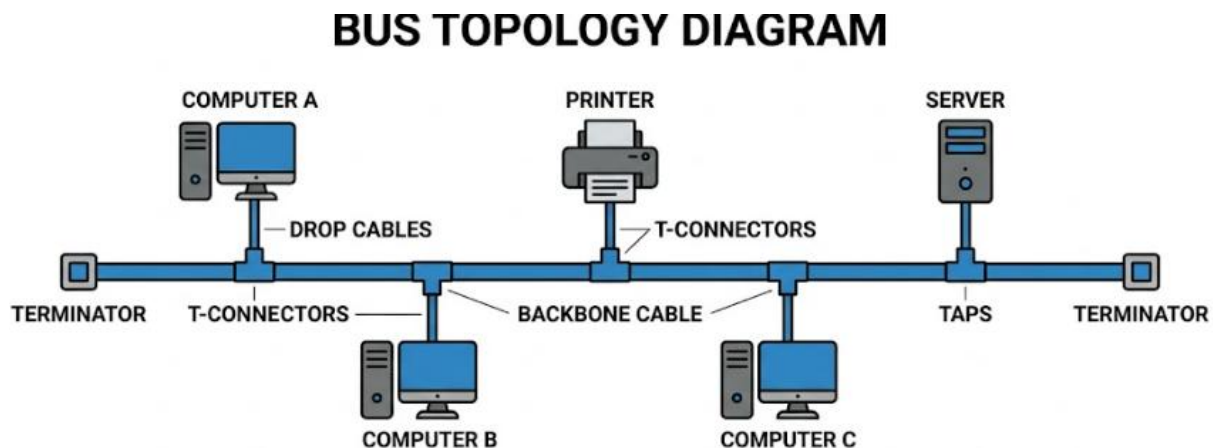


Day 2: Network Topologies Theory (1 hour):Types of network topologies: Bus, Star, Ring, Mesh. **Practical (1 hour):**Simulate a star topology using Cisco Packet Tracer.
Task: Draw a diagram of topologies with examples.

Task 1.

Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.



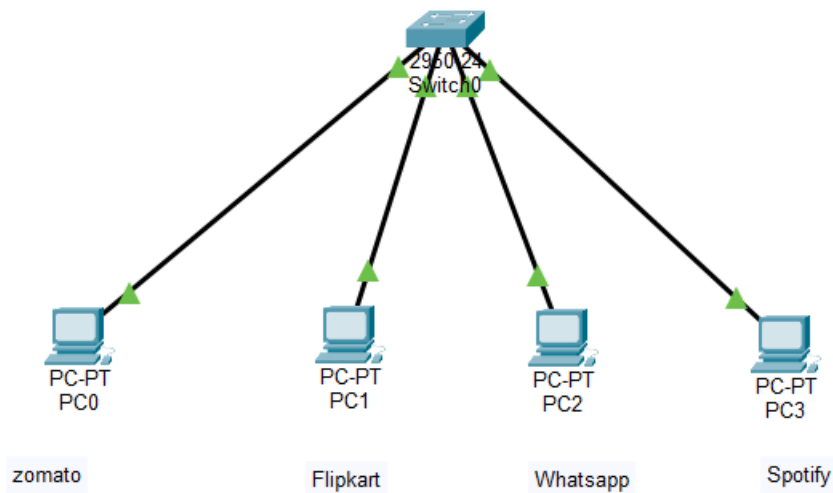
Key components:

- **Backbone cable:** Main cable through which data travels.
- **Devices/PCs:** Connected to the backbone using short connections.
- **Terminators:** Placed at both ends to prevent signal reflection.

example: Older Ethernet networks using 10BASE2/10BASE5 coaxial cable used bus topology. It was also commonly used in older small LAN installations.

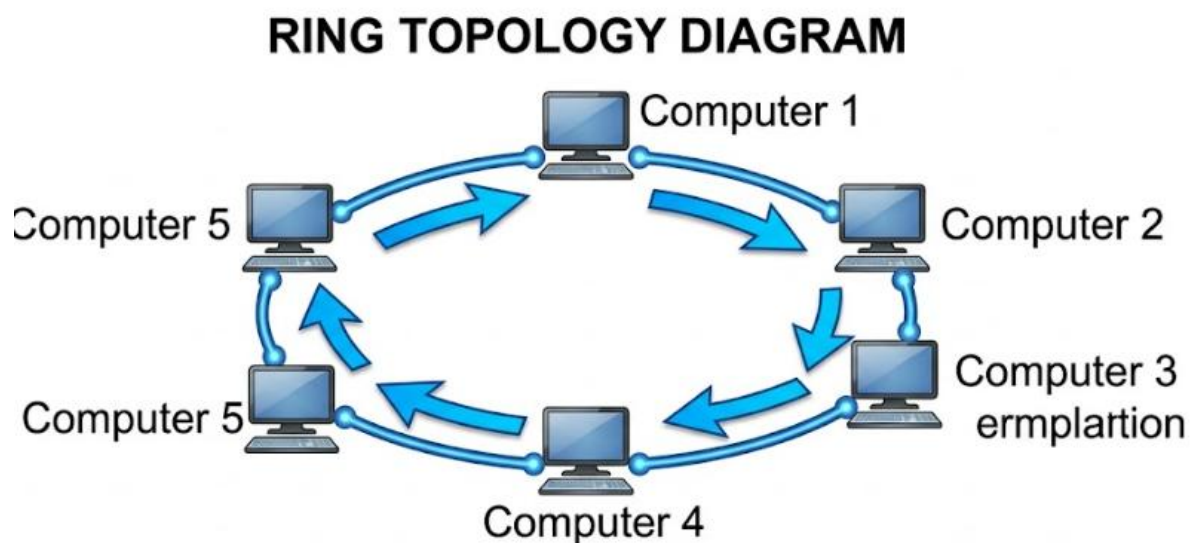
Task 2.

In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.



Task 3.

Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.
Hint: Use arrows to show the direction of data flow.



In a ring topology, data travels in one direction around the closed loop, from one device to the next. When a device sends data, the message passes through each intervening computer until it reaches the destination node. If an intermediate node finds the data is not intended for it, it re-transmits it to the next device in the ring.

Task 4.

Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Topology	Advantages	Disadvantages
Mesh	1. Very reliable because multiple paths are available.2. Failure of one connection usually does not bring down the entire network.	1. Requires a large amount of cabling and ports.2. Expensive and difficult to install and maintain.
Star	1. Easy to install, manage, and troubleshoot.2. Failure of one cable/device usually affects only that device.	1. If the central switch/hub fails, the whole network can be affected.2. Requires more cable than a bus topology.